

# The BAW Ising Machine as a Resonance Computer

Vadde et al. (2026) in the light of Docs. 173, 328 and 343

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# **The BAW Ising Machine as a Resonance Computer**

### Abstract

Vadde, Ovcharov, González, Khymyn, Litvinenko and Åkerman (Commun. Phys. 9:290, 8 September 2026, Open Access) present a time-multiplexed Ising machine with 2048 spins on two quartz bulk acoustic wave delay lines. This document reads the work as a case study of the FFGFT classification of resonance computers (Docs. 173, 328, 343): spins as phase states of a ring oscillator, the Barkhausen condition as winding quantisation, parametric  $\mathbb{Z}_2$  selection, a coupling regime with a chaos threshold, and — most clearly in quantitative terms — a resolution floor: 99.9 % approximate solutions are found reliably, exact solutions ( $E = 0$ ) are not as problem size grows. All quotations verbatim **[Q]**. Status of the classifications: **[S]**.

## Status Markers

|                     |                                                        |
|---------------------|--------------------------------------------------------|
| <b>[B]</b>          | Algebraically proved.                                  |
| <b>[K]</b>          | Numerically confirmed.                                 |
| <b>[S]</b>          | Structurally plausible; physical identification open.  |
| <b>[Q]</b>          | Taken from the source (Vadde et al.), quoted verbatim. |
| <b>[ASSUMPTION]</b> | Axiomatic postulate.                                   |

All verbatim quotations are from Vadde et al. (2026), Open Access under CC-BY 4.0; section references follow the structure of that paper.

## 1 The Source: What Vadde et al. Build

### Architecture

**[Q]**“We use two serially connected 20.5 MHz, 707  $\mu$ s bulk acoustic wave delay lines supporting 2048 spins. Our design provides all-to-all connectivity with 15-bit coupling resolution and finds approximate MAX-CUT solutions in 341 ms.” (Abstract)

The spins are not particles but phases of circulating RF pulses:

**[Q]**“The spin states +1 and −1 are encoded in the relative phase of the pulses, with bistable phases achieved by phase-binarizing the propagating RF pulses using a phase-sensitive amplifier (PSA).” (Section *Design*)

### The ring oscillator and the Barkhausen condition

**[Q]**“The BAW delay line, together with linear amplifiers and the PSA, forms a multi-physics ring oscillator. The key step of the design is to establish a stable circulation of RF pulses within this loop. This requires satisfying the Barkhausen stability criteria: the loop gain must be larger than one, and the total phase shift around the loop must be an integer multiple of  $2\pi$ .” (Section *Design*)

### Parametric binarisation

**[Q]**“The PSA further narrows the stability criteria to only those signals with either phase 0 or  $\pi$  relative to the pumping signal that is twice the reference frequency.” (Section *Design*)

### Minimum cycles and pulse duration

**[Q]**“Due to the resonance mechanism underlying phase-sensitive amplification, a minimum of 5–10 cycles per pulse is required, setting a lower bound on the pulse duration to  $\sim 250$ –500 ns. This constraint dominates and effectively determines the pulse period.” (Section *Design*)

## Coupling strength and chaos threshold

**[Q]**“The amplitude of the coupling signal can be tuned using the potentiometer of the DAC, allowing an additional control to ensure the injected signal remains within 5–30 % of the circulating RF signal and does not overpower it, avoiding chaotization of the oscillatory circuit.” (Section *Design*)

**[Q]**“The performance of BAWIM depends on the coupling strength, and there exists an optimal coupling strength depending on the problem instance.” (Section *Performance*)

## Approximation versus exactness

MAX-CUT:

**[Q]**“Here, we observe that BAWIM attains up to 95 % of HbSB’s best performance in terms of Ising energy, and up to 99.99 % in terms of the MAX-CUT score.” (Section *Dependence on the edge density*)

Number partitioning:

**[Q]**“Here, the exact solution corresponds to a difference of 0, while the 99.9 % approximate solution indicates that the difference is less than 0.1 % of the maximum possible value [...]. In all cases, BAWIM outperforms the HbSB algorithm, achieving a 100 % success rate in most instances when considering the 99.9 % approximate solution.” (Section *Number partitioning*)

Figure 5(b) shows what the text does not spell out: the curve “BAWIM Exact” falls from about 90 % success at 32 numbers to near zero at 2048 numbers, while “BAWIM 99.9 %” stays at 100 % across the whole range.

Sudoku:

**[Q]**“Notably, even when the Ising energy approaches the ground state, the Sudoku solution remains far from correct. [...] Sudoku requires exact constraint satisfaction; therefore, a near-ground-state Ising energy may still correspond to an invalid solution.” (Section *Sudoku*)

## Thermal stability

**[Q]**“In contrast, our BAWIM exhibits a temperature coefficient of phase accumulation of only 780 deg/°C, which is  $2.21 \times 10^4$  times lower than that of the CIM.” (Section *Comparison with CIMs*)

## 2 Classification 1: Ising Machine = Resonator (Doc. 173)

Doc. 173 §*The Arc Closes* says of the coherent Ising machine: “This resonator also has discrete states: it oscillates in either one or the other phase — spin-up or spin-down, zero or one. The excitation is analogue — continuous laser light, continuously adjustable coupling strengths. The state is discrete.” And in the conclusion: “An Ising machine is a resonator. All discrete from inside. All analogue from outside.”

BAWIM is this description in its purest form — no photonics, no cryogenics. The states are phases 0 and  $\pi$ ; the excitation is a continuous RF carrier; the coupling is an analogue current with 15-bit resolution. The spin is not an object but a phase relation. **[S]**

### 3 Classification 2: Barkhausen as Winding Quantisation (Doc. 314)

The Barkhausen condition “integer multiple of  $2\pi$ ” is a winding number: only states whose accumulated phase around the closed loop is  $2\pi n$ ,  $n \in \mathbb{Z}$ , circulate stably. **[B]** (Doc. 314, Ch. D2): “winding numbers are topological”; the mode structure on  $T^4$  is fixed by integer winding around every cycle.

The ring oscillator is an  $S^1$  — a single cycle. The Barkhausen condition is winding quantisation on that  $S^1$ . BAWIM thus realises technically what FFGFT posits for  $T^4$ : states exist only on integer windings. **[S]**

### 4 Classification 3: Parametric $\mathbb{Z}_2$ Selection (Doc. 339)

The PSA is pumped at  $2\omega$  and passes only phases 0 and  $\pi$ . This is period doubling: the phase state space  $S^1$  is reduced to two points  $\{0, \pi\} \cong \{+1, -1\}$ .

**[B]** (Doc. 339, §3.1): The fixed points  $\{+1, -1\}$  of the Frobenius on  $GF(27)^*$  form the massive  $\mathbb{Z}_2$  sector. **[B]** (Doc. 336): The norm classes  $\{+1, -1\}$  separate light from heavy.

BAWIM’s parametric selection is a technical realisation of the same dichotomy: from a continuous phase space exactly two fixed points are filtered out. That there are two and not three is due to pumping at  $2\omega$ ; pumping at  $3\omega$  would yield a  $\mathbb{Z}_3$  selection  $\{0, 2\pi/3, 4\pi/3\}$  — the Potts machine that Whitehead et al. (Ref. 12 in Vadde et al.) built in CMOS. **[S]**

### 5 Classification 4: Coupling Regime with Chaos Threshold (Doc. 328)

Doc. 328 treats coupling regimes of coupled oscillators: the Adler model, synchronisation, phase locking, and the limit beyond which coupling destroys the oscillator. BAWIM sets exactly this limit operationally: injection at 5–30 % of the circulating signal, “avoiding chaotization”.

The existence of an *optimal* coupling strength per problem instance (Suppl. Note 5) is the Adler sweet spot: too weak — no synchronisation, the system stays in random states; too strong — chaotisation, phase binarisation breaks down. In between lies the window in which the oscillator bank settles into the energy minimum. **[S]**

### 6 Classification 5: The Resolution Floor (Doc. 343 §G’)

This is the classification with the clearest quantitative support. Doc. 343 §G’ states: a resonance-based computer is capped at  $N_{\max} \approx 1/\varepsilon$ , where  $\varepsilon$  is the coupling tolerance; it finds minima to tolerance but not the exact zero. **[B]**

BAWIM displays this behaviour in three forms:

**MAX-CUT.** 95 % of Ising energy, 99.99 % of cut value. The optimum is reached to within fractions of a percent — but not exactly.

**Number partitioning.** The difference  $E = |\sum_A a_i - \sum_B a_i|$  is brought below 0.1 % of the total sum with 100 % success. The *exact* partition  $E = 0$ , by contrast, is still found in ~90 % of runs at 32 numbers and practically never at 2048 numbers (Fig. 5b). Coupling resolution is 15 bit; the integers are  $\leq 178$ ; the total sum at 2048 numbers is  $\sim 1.8 \times 10^5$ . A tolerance of

0.1 % corresponds to  $\sim 180$  — the magnitude of a single number. The machine solves the problem to within one element, not to zero.

**Sudoku.** At 99.42 % of ground-state energy the solution is still invalid. Sudoku is a constraint-satisfaction problem: there is no “almost right”. Vadde et al. say so themselves: “a near-ground-state Ising energy may still correspond to an invalid solution.” That BAWIM nevertheless solves the Sudoku is due to the smallness of the problem (729 spins) and to the ground-state energy being an isolated minimum here; HbSB fails on the same problem.

**The classification.** All three findings are consistent with Doc. 343: the resonance computer is an analogue computer with a resolution floor. It is *not* worse than a digital algorithm — on approximations it is better (NPP 99.9 %: BAWIM beats HbSB throughout). It is *different*: it finds the neighbourhood of the minimum, not the point. For optimisation that is the right task; for exact arithmetic it is not. **[S]**

## 7 What Does Not Correspond

**Classical, not quantum.** BAWIM is a classical RF ring oscillator at room temperature. It claims no quantum advantage and asserts none. The FFGFT classifications concern resonator structure, not quantum coherence.

**One cycle, not four.** The ring oscillator is  $S^1$ ;  $T^4$  has four independent cycles. The winding parallel (Classification 2) holds for one cycle; the  $T^4$ -specific statements (fixed points,  $\mathbb{Z}_3$  triality, mass assignment) have no counterpart in BAWIM.

**Benchmark caveat.** Vadde et al. stress twice that no parameter optimisation was performed for either HbSB or BAWIM. The quantitative comparisons (341 ms versus 102 ms; NPP success rates) are therefore snapshots. The *qualitative* structure — approximation reliable, exactness scale-dependent — is unaffected.

**The resolution floor here is technical, not fundamental.** The 15-bit coupling resolution is an electronics choice (AD9708 DAC), not a law of nature. Doc. 343 §G' speaks of  $\varepsilon = \xi$  as the natural floor; BAWIM's  $2^{-15} \approx 3 \times 10^{-5}$  happens to lie in the same order of magnitude as  $\xi = 1.33 \times 10^{-4}$ , without any connection being claimed here.

## 8 Summary

| BAWIM (Vadde et al.)            | FFGFT                                             | Doc.     | Status     |
|---------------------------------|---------------------------------------------------|----------|------------|
| spin = phase $0/\pi$ of a pulse | Ising machine = resonator                         | 173      | <b>[S]</b> |
| Barkhausen: phase = $2\pi n$    | winding quantisation on $S^1$                     | 314      | <b>[S]</b> |
| PSA at $2\omega$ : $\{0, \pi\}$ | $\mathbb{Z}_2$ fixed points $\{+1, -1\}$          | 339, 336 | <b>[S]</b> |
| coupling 5–30 %, chaos limit    | Adler regime, coupling threshold                  | 328      | <b>[S]</b> |
| 99.9 % yes, exact no            | resolution floor $N_{\max} \approx 1/\varepsilon$ | 343 §G'  | <b>[S]</b> |



BAWIM is the clearest published case to date of a resonance computer in the sense of Docs. 173/343: thermally stable, without post-selection, without cryogenics, with reproducible approximate solutions and a measurable resolution floor for exact ones. It confirms the FFGFT classification not as a theory but as a taxonomy: what resonates finds minima — and finds them to tolerance.

**Source.** Vadde, V.; Ovcharov, R.; González, V. H.; Khymyn, R.; Litvinenko, A.; Åkerman, J.: *A 2048-spin bulk acoustic wave Ising machine for number partitioning and Sudoku*. Commun. Phys. **9**, 290 (2026). DOI: 10.1038/s42005-026-02846-7. Open Access, CC-BY 4.0. Received 2 March 2026, accepted 20 August 2026, published 8 September 2026.